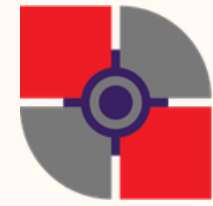




**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA



**PERSAKA**  
PERSATUAN MAHASISWA SAINS KOMPUTER

# IMPACT REPORT



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UTMxHackathon'26

**Date:**  
24 May 2026

**Venue:**  
N28a, Faculty of Computing, UTM Skudai

<https://utm-hackathon.vercel.app>

# What is UTMXHACKATHON?



UTMxHACKATHON'26 is an annual innovation event organised by PERSAKA. It is designed to challenge university students to move beyond theoretical frameworks and develop scalable solutions for authentic industry problem statements.

**The Vision:** To become the ultimate platform for industry-led student innovation, connecting top-tier technical talents with the corporate world.





# Event Timeline

**9 March - 3 May 2026**

Workshop  
(Online)

**4 May - 10 May 2026**

Preliminary Round  
(Online)

**23 May 2026**

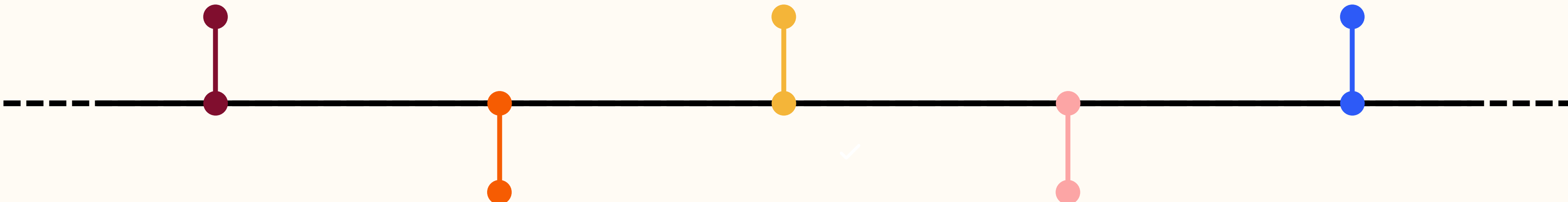
Finals Round  
*(Exhibition & Pitching)*  
(Physical)

**4 May 2026**

Opening Ceremony &  
Participant Briefing  
(Online)

**18 May - 23 May 2026**

Finals Round  
(Online)



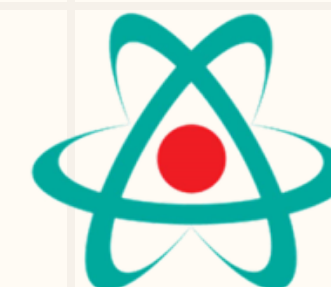


<https://utm-hackathon.vercel.app>

# OUR 2026 REACH

## Breakdown:

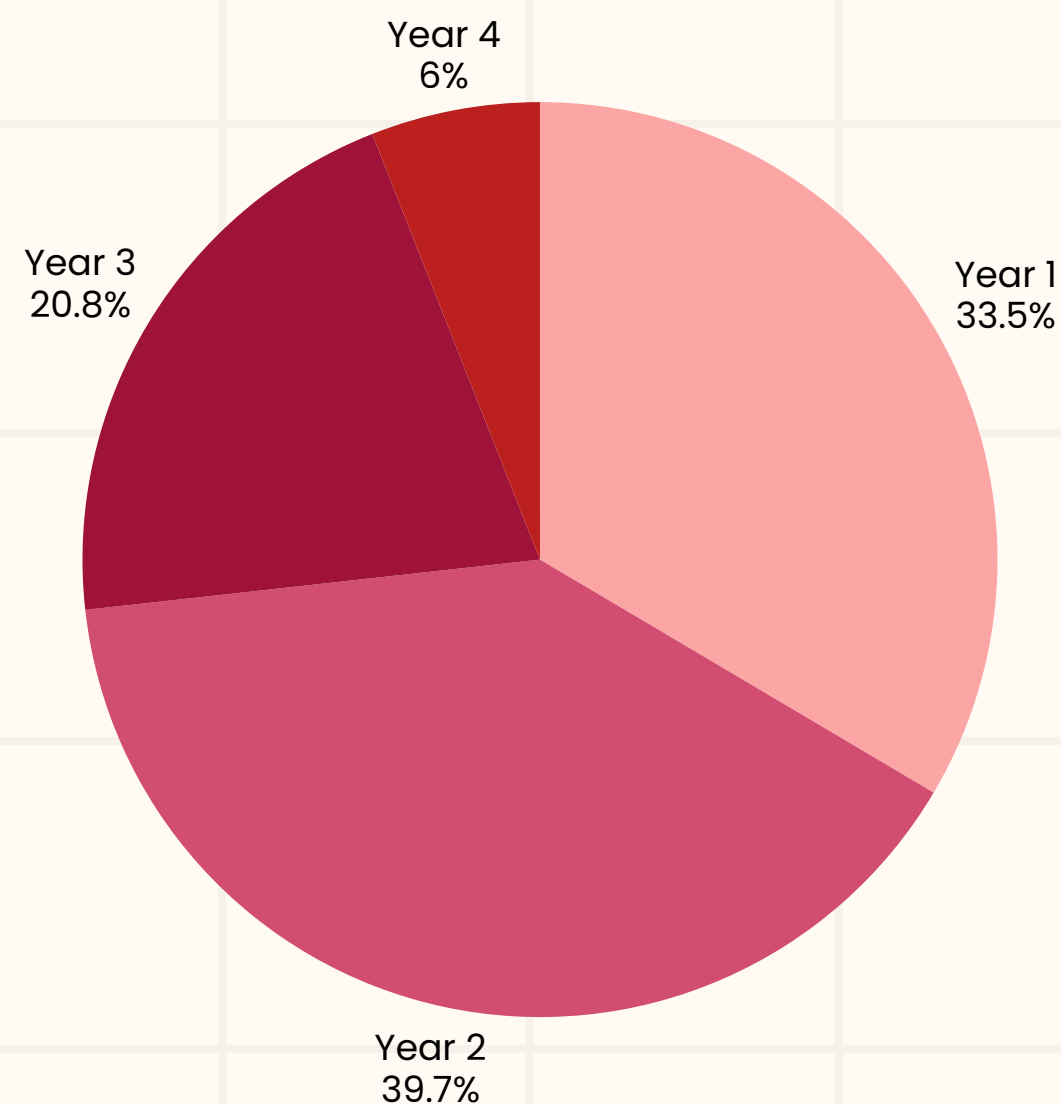
123 Teams with  
399 Participants  
from 24 Universities  
across Malaysia





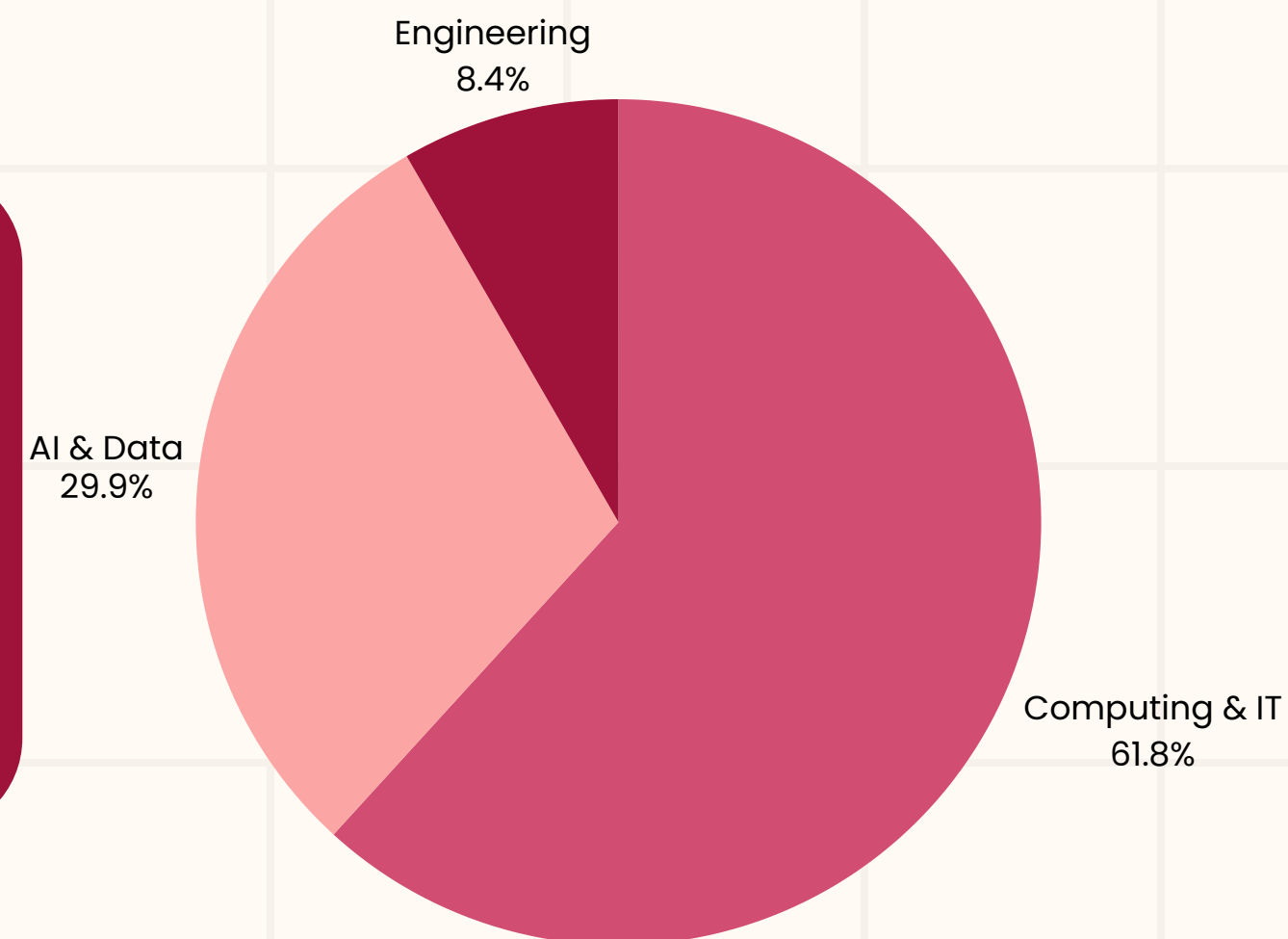
# Participant Demographic

**Year of Study**



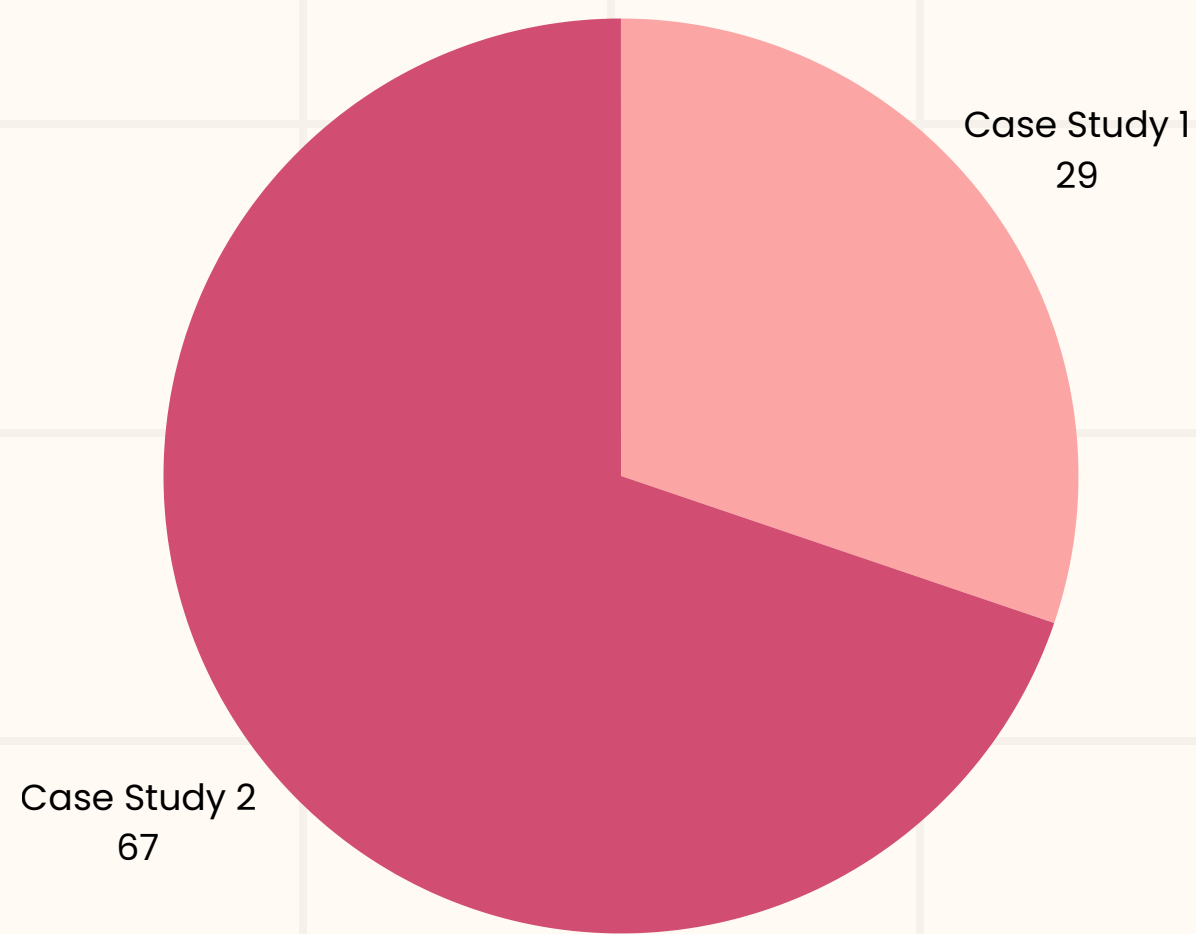
  
**399 Participants  
of Diverse Skillsets &  
Background**

**Course Breakdown**



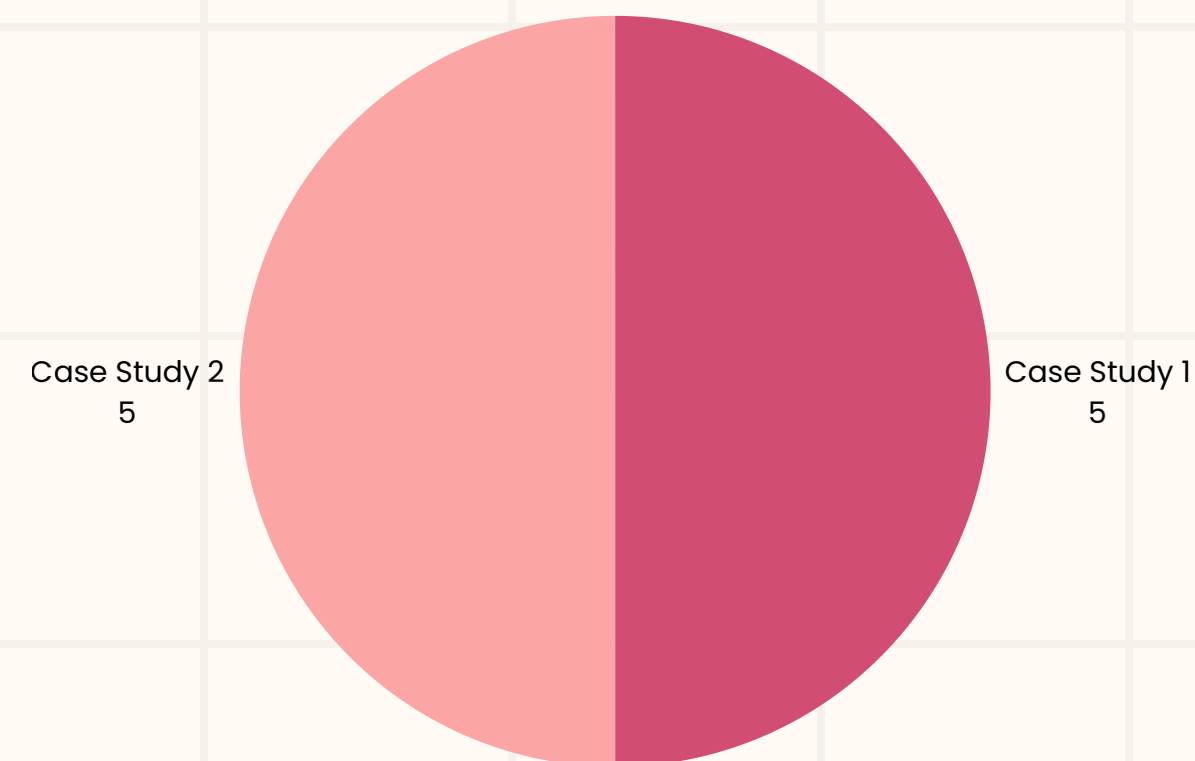
# PRELIMINARY ROUND

Preliminary Submission Breakdown



  
**96 Submissions**  
10 Finalists

Finalists Breakdown



# FINALISTS SHOWCASE

Let's see the innovative solution by the  
TOP 10 finalists of UTMxHackathon'26.







# 1. MoneyKawKaw by Protect Liver



An AI-powered active financial companion integrated within GXBank designed to transform traditional passive banking into an active financial habit builder.

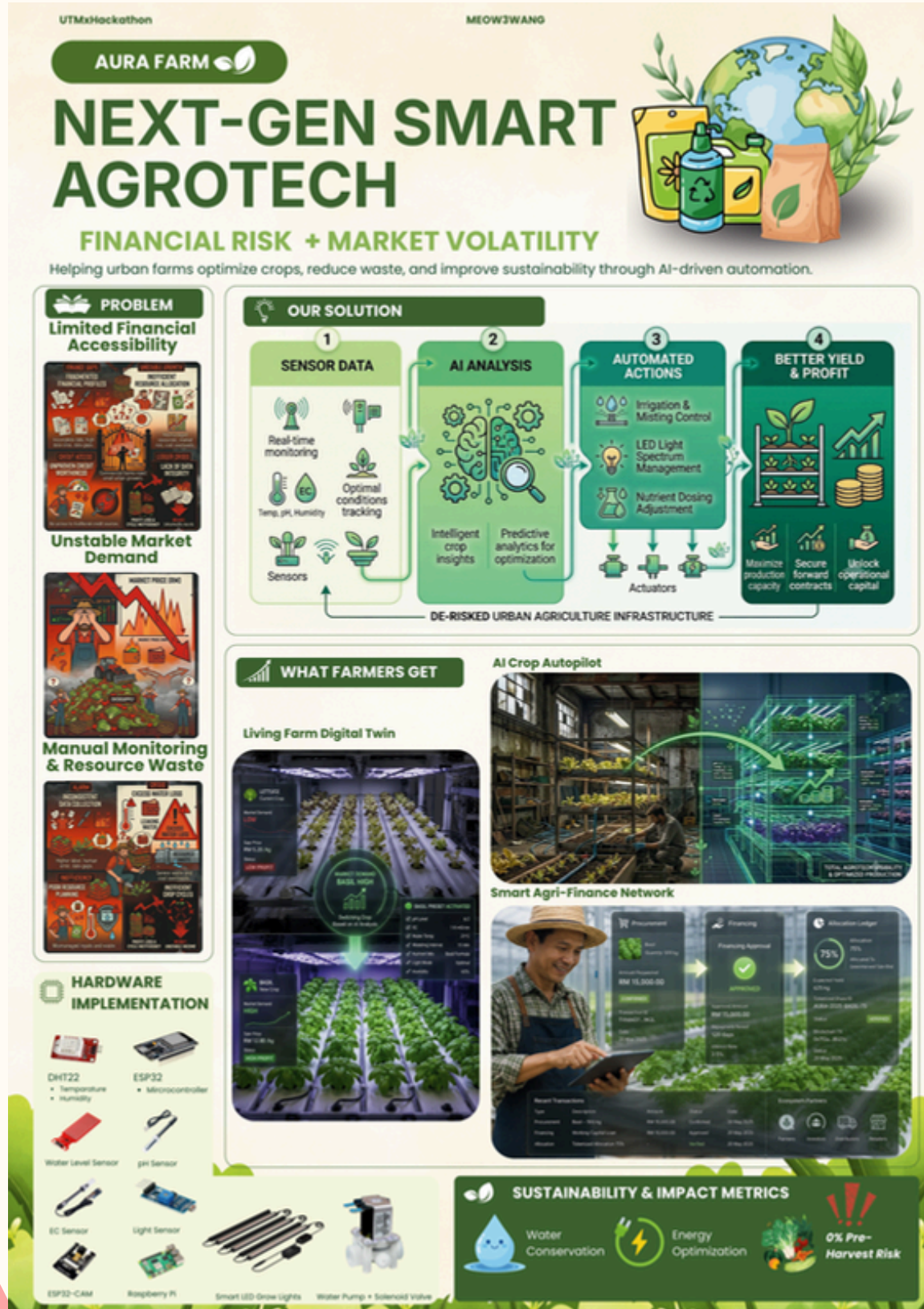
[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)







## 2. AuraFarm by MEOW3WANG



Next-generation smart agrotech and Web3 agri-finance platform designed to help urban farms optimize crop yields, eliminate market price volatility, and unlock operational capital through on-chain escrow tokens and AI-driven automation.

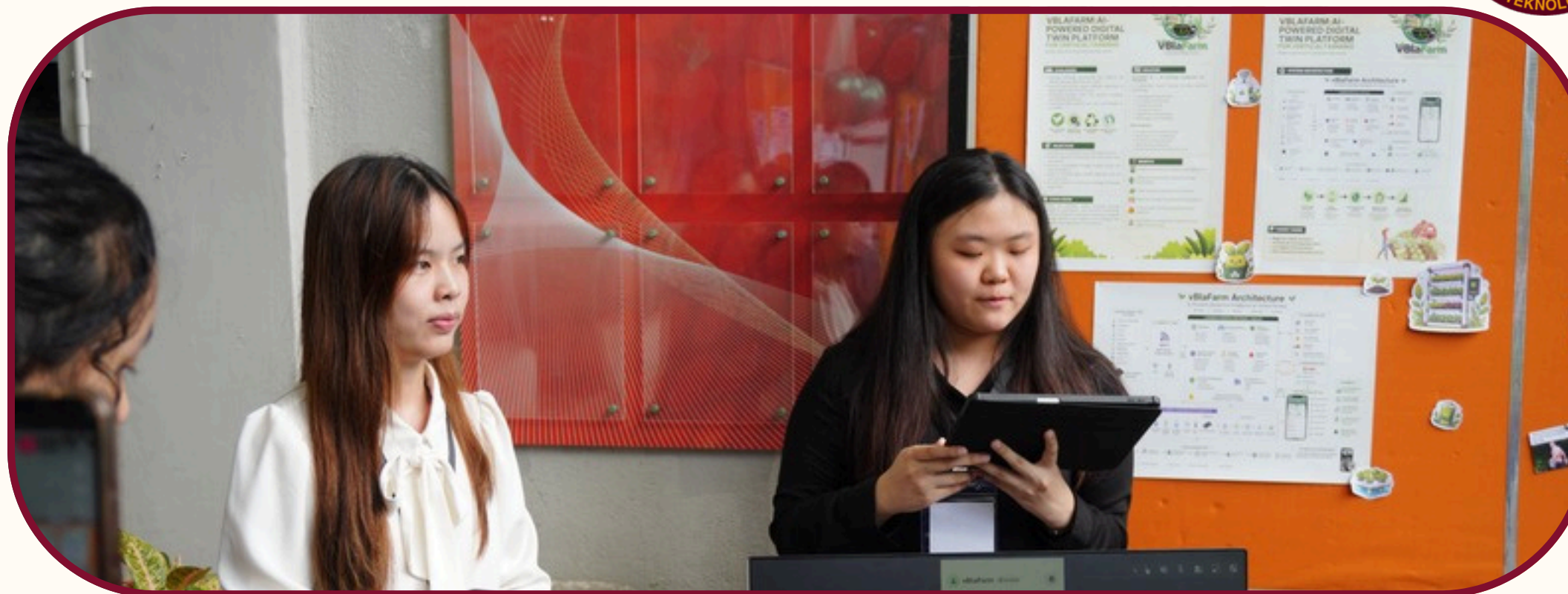
[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)



### 3. vBlaFarm by Blablabla



An AI-powered operational intelligence platform using digital twins and augmented reality to help indoor vertical farms predict anomalies, automate workflows, and optimize crop yields.

[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)



**vBlaFarm**  
GROW SMARTER, HARVEST BETTER

**TEAM BLABLABLA**

**VBLAFARM: AI-POWERED DIGITAL TWIN PLATFORM FOR VERTICAL FARMING**

Modern Agriculture for Sustainable Yield Growth.

**CHALLENGES**

- Complex farming dashboards are difficult for elderly and low digital literacy users.
- Manual monitoring causes delayed responses to plant and environmental issues.
- Managing multiple racks and sensors increases operational complexity.
- Traditional alert systems are not user-friendly or accessible.

**OBJECTIVES**

- Develop an AI-powered smart farming assistant.
- Enable real-time monitoring and automated farm operations.
- Improve accessibility through simple, visual, and voice-friendly UI.
- Support AI-based plant health detection and AR rack scanning.
- Provide familiar AI assistance through WhatsApp integration.

**CONCLUSION**

FarmPilot AI simplifies smart farming through accessible AI, automation, and real-time monitoring. The platform improves operational efficiency, reduces technology barriers, and enables more inclusive and sustainable smart agriculture for everyone.

**SOLUTION**

FarmPilot AI — AI Farming Companion for Everyone

A mobile-first smart vertical farming platform combining:

- AI & Agentic Automation
- IoT Sensor Monitoring
- ARRack Scanning
- Computer Vision
- Digital Twin Visualization
- WhatsApp AI Accessibility

**Key Features**

- AI Operational Control Center
- Real-Time Farm Dashboard
- AI Chat Farming Assistant
- Plant Disease Detection
- Smart Automation & Safety Controls

**BENEFITS**

- Predictive operational insights using Agentic AI.
- Reduced monitoring complexity through automation.
- Faster anomaly detection and response.
- Beginner-friendly AI workflows and guidance.
- Unified smart farming operations in one platform.
- Accessible AI assistance for elderly and low digital literacy users.

Eco-Friendly Process   Real Time Monitoring   Sustainable Impact   Better for Our Planet



## 4. ThinkTwice by kamihack

KamiHack

**ThinkTwice**  
An AI-powered Financial Behavioural Intervention System.

Try Our Demo

**The Real Problem**

- 1 Too Late**  
Only realize that you overspent after the money is gone.
- 2 Passive Tracking**  
Most banking apps observe behaviour but never intervene.
- 3 Missing Link**  
Awareness alone does not change habits.

**Core Features**

- AI Agentic Intelligence**  
Multi-agent AI architecture instead of relying on a single AI model.
- Real-Time Nudges**  
Instead of waiting for monthly reports, ThinkTwice reacts instantly.
- Smart Savings Radar**  
AI-powered location intelligence for cheaper alternatives, and verified local deals.
- Gamification**  
Gamifying discipline through metric scores, progress streaks, rewards, and leaderboards.
- Analytics**  
Tracks: Savings, Resilience Scores, avoided overspending, Smart Radar data, streaks.

**Tech Stack**

- Frontend: Flutter
- Backend: Node.js Express
- Database: Firebase Authentication, Firestore
- Maps: Google Maps API
- AI: Gemini AI

**Our Solution**

- Proactive Nudging**  
AI detects risk before the habit forms.
- Smart Radar**  
Intercepts expensive choices with cheaper, community-verified local routes.
- Gamified Resilience**  
Turn saving into a game: personalized buddy, scores, streaks, and leaderboards.

| Traditional Banking    | ThinkTwice                     |
|------------------------|--------------------------------|
| Detects AFTER Spending | Predicts BEFORE Spending       |
| Passive Tracking       | Active Intervention            |
| Monthly Analytics      | Real-time Behavioural Guidance |

**How It Works**

- 01 Detect**  
AI analyses financial behaviour and spending risk
- 02 Nudge**  
Immediate intervention triggered
- 03 Act**  
User takes financial action
- 04 Reward**  
Positive behaviour reinforced
- 05 Repeat**  
Learning loop improves future recommendations



Transforms passive banking into an intelligent behavioural guidance ecosystem that helps users make smarter financial decisions in the moment.

[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)



# 5. Finora by No Sleep Till Demo



AI-powered financial resilience platform designed to help Malaysian tertiary students and early-career graduates develop sustainable financial habits through behavioural economics, AI-driven coaching, and automated financial guidance.

[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)



**UTM X HACKATHON 2026**

**F. FINORA**

**AI-POWERED FINANCIAL RESILIENCE BANKING APP**

**TEAM NO SLEEP TILL DEMO**

**BUILT FOR MALAYSIAN YOUTH**

Students Fresh Graduates Early Career Professionals Future Achievers

**THE PROBLEM**

Malaysian youth are facing a financial resilience crisis.

- Debt Traps & BNPL Dependency
- Impulsive Spending
- Lack of Savings Discipline
- Financial Anxiety
- Reactive Financial Behaviour

**WHY IT MATTERS**

- Proactive Debt Prevention**  
Avoid financial danger before it becomes a problem.
- Stronger Savings Habits**  
Build automatic savings and long-term wealth.
- Reduced Financial Stress**  
Improve financial confidence and mental well-being.
- Financial Discipline**  
Make smarter daily decisions with AI guidance.
- Long-Term Resilience**  
Build a secure financial future step by step.

**Traditional Banking Apps**

- Only show transactions
- Passive dashboards
- No proactive intervention

**VS**

**Our AI-Powered Solution**

- Predict financial risk
- Guides behavior
- Automates healthy habits
- Provides personalized financial coding

**Financial DNA Personalization Planner**

**SMART AI Nudges**

**Debt Risk Warning System**

**Financial Health Score**

**Weekly Behavioural Challenges**

**Monthly Commitment Tracker**

**Financial Resilience Journey**

**Social Savings Benchmark**

**Cashflow status**

**FD Autopilot Summary**

**Recent Transaction Tracking**

**Mobile Banking-Style Navigation**

**4** DNA personality types

**12+** AI-powered features

**5** stages Financial resilience journey

**3-tier** Debt alert system

**0** Willpower needed

**HOW IT WORKS (SYSTEM ARCHITECTURE)**

Mobile App (iOS/Android) → Financial DNA Engine → Behavioral Analytics → AI Recommendation Engine → Smart Nudge Generator → Risk Detection Engine → Financial Health Scoring System → Savings Automation Layer → GBank Ecosystem Integration

**SCAN TO SEE THE APP IN ACTION**

**Build secure habits | Drive behavioural change | Achieve resilience**



UTMHACKATHON'26 • PERSAKA • UTM SKUDAI • 23 MAY 2026  
SeedDown POSTER 1 OF 4  
Track: Precision Urban Agriculture • Case Study 1

**From Guesswork to Closed-Loop Growing**

SeedDown is an AI + IoT vertical farming platform that senses the farm, decides what it needs, acts automatically, and visualizes the result.

**PROBLEM • VISION • SOLUTION**

Confused Grower → Unhealthy Crop → SeedDown Sensors → AI Decides → Healthy Farm → Confident Grower

73% Urban farmers quit 8+ Parameters to monitor

Most small growers fail because they cannot continuously monitor conditions, tune crop thresholds, or respond fast enough when something goes wrong.

**WHY URBAN FARMERS GIVE UP**

- Overwhelmed by Complexity**  
8+ growing parameters (temp, humidity, pH, EC, CO<sub>2</sub>, soil, light, gas) are impossible to monitor manually around the clock.
- Response Time is Too Late**  
By the time a problem is visible, the plant is already stressed or diseased. Manual checks cannot catch issues in time.
- Zero Guidance for Beginners**  
New growers do not know the right thresholds for their crop, package, or growing goal. There is no feedback loop to learn from.
- Commercial Farms Cannot Scale**  
Energy waste, poor zone visibility, manual labour, and no automation make vertical farming expensive to scale for operators.

**SEEDDOWN CLOSED LOOP**

- Sense**  
ESP32 reads temperature, humidity, soil, light, pH, EC, CO<sub>2</sub>, gas, water level
- Transmit**  
Readings sent via HTTPS to Express API → stored in Firebase Firestore
- Decide**  
AI + rule engine compares readings with thresholds, generates commands
- Act**  
WATER\_ON / LIGHT\_ON / FAN\_ON / BUZZER\_ON / alerts sent to ESP32 actuators
- Visualize**  
Live dashboard + 3D Digital Twin shows farm status in real time

Loop repeats continuously • Condition-based, not always-on

**TWO MODES, ONE PLATFORM**

- Beginner Mode**  
HOME GROWER
  - QR scan to onboard
  - Photo-assisted setup
  - Goal-based AI thresholds
  - 3D farm preview
  - Simple alerts & advice
  - Community support
- Commercial Mode**  
FARM OPERATOR
  - Multi-zone dashboard
  - Farm Master + Zone Nodes
  - Disease analysis
  - What-If planning
  - 3D Digital Twin
  - Energy & profit analytics

**SYSTEM DATA FLOW**

QR Package (Scan to register) → ESP32 Node (Sensor readings) → Node.js API (Ingestion & logic) → Firebase (Firestore DB) → AI + Rules (Logic / Control) → Dashboard (3D Twin) → Actuator (Command sent)

End-to-end IoT loop • ESP32 + Cloud + AI + Dashboard + ESP32

Team: next level utm  
Wong Jia Hui • Lee Mei Shuei • Loh Su Ting • Christ Ting Shin Ling • Wong Zi Qi

UTMHACKATHON'26 • 23 May 2026 • UTM Skudai

## 6. SeedDown by next level utm



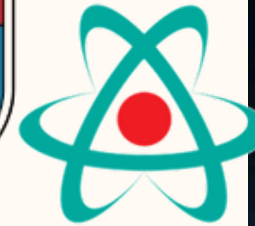
An AI-powered closed-loop farm operations platform using 3D digital twins and IoT automation to help urban growers and vertical farms diagnose crop risks, optimize resource usage, and safely automate agricultural workflows.

[Demo Link](#)

[Presentation Link](#)



# 7. VertiSync by John Hyrax



An AI-driven, IoT-integrated vertical farming management platform designed to optimize crop yield, automate environmental controls, and minimize resource consumption.

[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)

## VERTISYNC

The Intelligent Software Brain for Precision Indoor Agriculture

- Real-Time Telemetry Pipeline**  
Instant multi-sensor tracking via WebSockets.
- Predictive AI Brain**  
Advanced evaporation and crop stress trajectory modeling.
- Autonomous Actuation**  
Direct edge-node command relays for fans and irrigation.
- Scalable Crop Library**  
One-click automated environmental recipe deployment.



## 8. EcoPal by Avengers



A gamified personal finance app built with Flutter that turns your financial habits into a living, breathing garden. Instead of staring at dry charts, you tend to plants that grow or wilt based on how well you manage your money — with an AI-powered pet companion guiding you every step of the way.

[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)



## 9. VertiFlow by mhmdsamer's Terminal



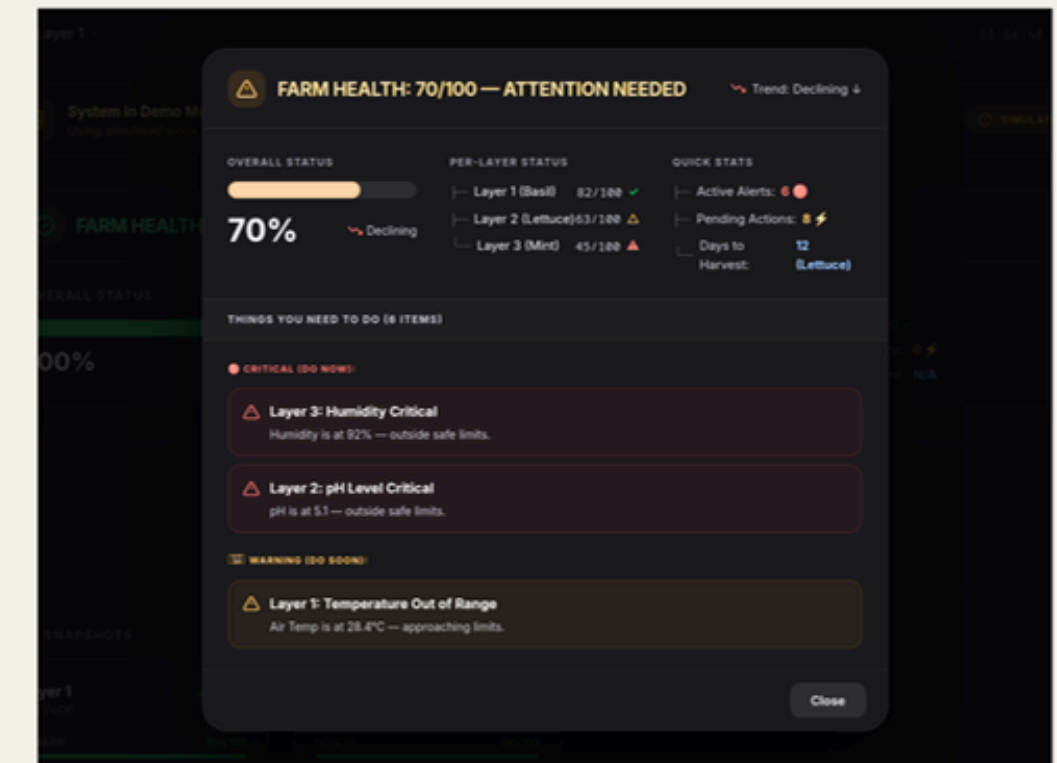
An IoT-driven vertical farming platform that leverages real-time monitoring and advanced analytics to automate workflows, trigger intelligent alerts, and ensure highly efficient, predictable crop growth.

[Github Repo](#)

[Demo Link](#)



### PROJECT INTRODUCTION



#### KEY FEATURES

- Real-time IoT monitoring
- Smart automation rules (no coding)
- Grow Cycle Management
- Multi-zone tracking
- 24/7 alerts

#### Target Audience

- 200,000+ urban farms in SE Asia
- Eight Agriculture's Faculty in Malaysia
- Small Agriculture Businesses



## 10. GX.AI by let me solo it



### GX.AI Context Engine

Intelligent banking. Automated habits.  
Real-time insights.

**THE PROBLEM** **61%**  
of Malaysians struggle coming up with  
RM 1000 in case of an emergency  
"They only show what you've already lost"

Scan for Github Repo



**FEATURED**

**Velocity Nudges**  
Detects high weekend spending and diverts micro-funds to an emergency pocket

**Subscription Audits**  
Identifies dormant subscriptions and re-routes wasted money to High-Yield Pockets

**Salary Routing**  
Instantly processes salary inflows using the 50/30/20 rule

**Nerd Section**

 Next.js
  Tailwind CSS
  Framer Motion
  Anthropic

Muhammad Ikhwan | Let Me Solo It | UTM Shah Alam

An autonomous financial resilience dashboard for GXBank that uses real-time LLM analysis to track spending velocity and proactively intervene before financial stress occurs.

[Github Repo](#)

[Demo Link](#)

[Presentation Link](#)

# Participant Testimonials



UTMxHackathon'26 **allowed me to apply classroom concepts to real-world problems.** Instead of theoretical assignments, building a practical, user-focused solution showed me exactly how technical knowledge drives value in the industry.

- **Teo Jing Ying (MEOW3WANG), UM, Software Engineering**

As a solo developer, UTMxHackathon was a tough but amazing experience that **pushed me to build and pitch** the 'GX.AI Context Engine' from scratch. The organisers were incredibly helpful and welcoming to non-UTM students. I will 100% join the next one!

- **Muhammad Ikhwan (Let Me Solo It), UiTM, Intelligent System**

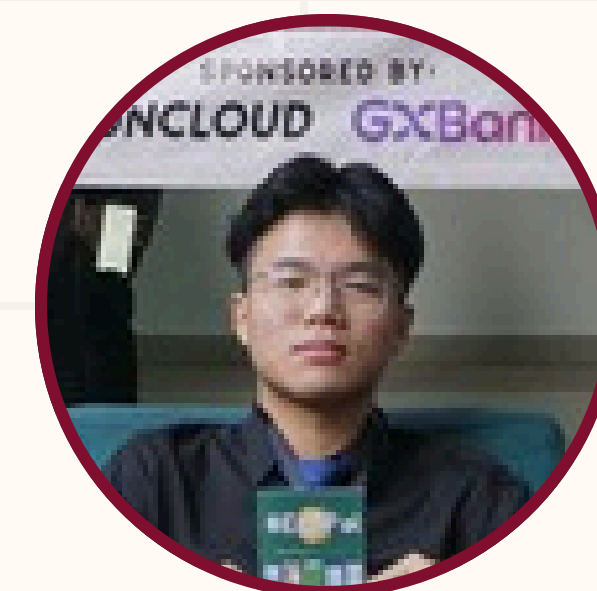




# Participant Testimonials

A wonderful experience with fantastic management! The broad case study **pushed us to think outside the box** and develop a truly creative solution. From the competition to the judges and the organisation, everything was great. I definitely wish to participate again!

- **Gan Rui En (Avengers), UTM, Software Engineering**



UTMxHackathon'26 pushed us right past our limits. Building under intense pressure taught us to rapidly **bridge the gap between hardware and software**. The invaluable feedback from the judges made it a deeply rewarding weekend that truly accelerated our growth as developers!

- **Ng Wei Kean (John Hyrax), AIMST, Computer Science**

# Participant Testimonials



Thank you for organizing the hackathon! We invested a significant amount of time developing and refining the backend and technical components of our solution. Overall, we enjoyed participating, the flow of the finals was well-handled, and we truly appreciate the opportunity to learn and compete. Keep up the good work! 🙌

- (next level utm), UTM, Software Engineering

UTMxHackathon'26 was an amazing experience. We learned so much, from **refining ideas to solving challenges under pressure**. Being recognized with the Best Innovation & Idea Award and as a Top 10 Finalist was truly meaningful for our team!

- Kwek Chee Ling (kamihack), UM, Software Engineering





# Participant Testimonials

The most interesting challenge was **building within GXBank's existing ecosystem rather than starting from scratch**. We actively analyzed the app and realized their 'Pockets' feature was highly unique, using it as the foundation for our ideas. It was an incredible experience figuring out how to enhance a feature users already know!

- **Chiam Huai Ren (Protect Liver), UM, Data Science**



Facing a public pitch and Q&A with the judges had me so anxious I almost didn't want to do it, but I got up there handled it fine, and realized I was stressing over nothing. This hackathon helped me to **prove to myself I could handle the pressure easily**.

- **Samer Salem (mhmdsamer's Terminal), UTM, Software Engineering**

# The Road Ahead



## **Bigger Network, Bigger Challenges**

Opening the doors even wider to connect, compete, and collaborate with an even larger pool of top-tier tech talents from universities all across Malaysia.



## **Direct Pathways to Your Career**

Bring in more industry giants like GXBank, giving exclusive access to real-world challenges, direct mentorship, and fast-tracked internship opportunities.



## **Beyond the Weekend**

Post-hackathon pathways to turn winning prototypes into real-world startups or integrated industry solutions.





# Thank You!

You pushed your limits, bridged the gap between theory and reality, and built incredible solutions. The massive success of UTMxHackathon'26 is entirely driven by your passion, late nights, and relentless innovation.

The hackathon might be a wrap, but the building doesn't stop here. Follow PERSAKA's socials to stay in the loop for our next hackathons, tech workshops, and industry events!

**See you next year at UTMxHackathon'27!**



PERSAKA



PERSAKA UTM



PERSAKAUTM



UTMXHACKATHON26





# Pictures







# Pictures











# Pictures







# Pictures





# Closing Remarks



Thank you to everyone who made UTMxHackathon'26 our most impactful event yet. Seeing hundreds of students tackle real-world industry challenges head-on was incredible. We are deeply grateful to GXBank & Runcloud for empowering this generation of builders and innovators.

**Hoe Zhi Wan,**  
**Director,**  
**UTMxHackathon'26**

The energy during the physical pitching sessions was unmatched. Watching finalists rapidly debug and confidently present their prototypes was truly inspiring. A massive thank you to our dedicated activity committee and brilliant participants for making this fast-paced weekend a reality!

**Munif Baharum,**  
**Vice Director Activity,**  
**UTMxHackathon'26**

